



Graphs in Machine Learning

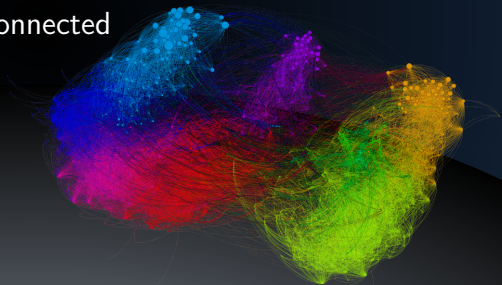
Similarity Graph Types

-neighborhood, k-NN, Fully Connected

Michal Valko

Isara Labs

Partially based on material by: Andreas Krause,
Branislav Kveton, Michael Kearns



Similarity Graphs

Similarity graph: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — **(un)weighted**

Task 1: For each pair i, j : define a **similarity function** s_{ij}

Task 2: Decide which edges to include

Similarity Graphs

Similarity graph: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — **(un)weighted**

Task 1: For each pair i, j : define a **similarity function** s_{ij}

Task 2: Decide which edges to include

ε -neighborhood graphs – connect the points with the distances smaller than ε

Similarity Graphs

Similarity graph: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — **(un)weighted**

Task 1: For each pair i, j : define a **similarity function** s_{ij}

Task 2: Decide which edges to include

ε -neighborhood graphs – connect the points with the distances smaller than ε

k -NN neighborhood graphs – take k nearest neighbors

Similarity Graphs

Similarity graph: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — **(un)weighted**

Task 1: For each pair i, j : define a **similarity function** s_{ij}

Task 2: Decide which edges to include

ε -neighborhood graphs – connect the points with the distances smaller than ε

k -NN neighborhood graphs – take k nearest neighbors

fully connected graphs - consider everything

Similarity Graphs

Similarity graph: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — **(un)weighted**

Task 1: For each pair i, j : define a **similarity function** s_{ij}

Task 2: Decide which edges to include

ε -neighborhood graphs – connect the points with the distances smaller than ε

k -NN neighborhood graphs – take k nearest neighbors

fully connected graphs – consider everything

Similarity Graphs

Similarity graph: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — **(un)weighted**

Task 1: For each pair i, j : define a **similarity function** s_{ij}

Task 2: Decide which edges to include

ε -neighborhood graphs – connect the points with the distances smaller than ε

k -NN neighborhood graphs – take k nearest neighbors

fully connected graphs – consider everything

This is art (not much theory exists).

http://www.informatik.uni-hamburg.de/ML/contents/people/luxburg/publications/Luxburg07_tutorial.pdf

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted
- **equivalent to: similarity function is at least ε**

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted
- equivalent to: similarity function is at least ε
- theory [Penrose, 1999]: $\varepsilon = ((\log N)/N)^d$ to guarantee connectivity N nodes, d dimension

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted
- equivalent to: similarity function is at least ε
- theory [Penrose, 1999]: $\varepsilon = ((\log N)/N)^d$ to guarantee connectivity N nodes, d dimension
- practice: choose ε as the length of the longest edge in the MST - minimum spanning tree

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted
- equivalent to: similarity function is at least ε
- theory [Penrose, 1999]: $\varepsilon = ((\log N)/N)^d$ to guarantee connectivity N nodes, d dimension
- practice: choose ε as the length of the longest edge in the MST - minimum spanning tree

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted
- equivalent to: similarity function is at least ε
- theory [Penrose, 1999]: $\varepsilon = ((\log N)/N)^d$ to guarantee connectivity N nodes, d dimension
- practice: choose ε as the length of the longest edge in the MST - minimum spanning tree

What could be the problem with this MST approach?

Similarity Graphs: ε -neighborhood graphs

Edges connect the points with the distances smaller than ε .

- distances are roughly on the same scale (ε)
- weights may not bring additional info $\rightarrow$ unweighted
- equivalent to: similarity function is at least ε
- theory [Penrose, 1999]: $\varepsilon = ((\log N)/N)^d$ to guarantee connectivity N nodes, d dimension
- practice: choose ε as the length of the longest edge in the MST - minimum spanning tree

What could be the problem with this MST approach?

Anomalies can make ε too large.

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?
- $k \approx \log N$ - suggested by asymptotics (practice: up to $\sqrt{N}$)

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?
- $k \approx \log N$ - suggested by asymptotics (practice: up to $\sqrt{N}$)
- for mutual k -NN we need to take larger k

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?
- $k \approx \log N$ - suggested by asymptotics (practice: up to $\sqrt{N}$)
- for mutual k -NN we need to take larger k
- mutual k -NN does not connect regions with different density

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?
- $k \approx \log N$ - suggested by asymptotics (practice: up to $\sqrt{N}$)
- for mutual k -NN we need to take larger k
- mutual k -NN does not connect regions with different density
- why don't we take $k = N - 1$?

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?
- $k \approx \log N$ - suggested by asymptotics (practice: up to $\sqrt{N}$)
- for mutual k -NN we need to take larger k
- mutual k -NN does not connect regions with different density
- why don't we take $k = N - 1$?
 - space and time

Similarity Graphs: k -nearest neighbors graphs

Edges connect each node to its k -nearest neighbors.

- asymmetric (or directed graph)
 - option OR: ignore the direction
 - option AND: include if we have both direction (mutual k -NN)
- how to choose k ?
- $k \approx \log N$ - suggested by asymptotics (practice: up to $\sqrt{N}$)
- for mutual k -NN we need to take larger k
- mutual k -NN does not connect regions with different density
- why don't we take $k = N - 1$?
 - space and time
 - manifold considerations (preserving local properties)

Similarity Graphs: Fully connected graphs

Edges connect everything.

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp \left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2} \right)$$

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp \left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2} \right)$$

- why the exponential decay with the distance?

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp\left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

- why the exponential decay with the distance?
- σ controls the width of the neighborhoods

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp\left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

- why the exponential decay with the distance?
- σ controls the width of the neighborhoods
 - similar role as ϵ

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp\left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

- why the exponential decay with the distance?
- σ controls the width of the neighborhoods
 - similar role as ε
 - a practical rule of thumb: 10% of the average empirical std

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp \left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2} \right)$$

- why the exponential decay with the distance?
- σ controls the width of the neighborhoods
 - similar role as ε
 - a practical rule of thumb: 10% of the average empirical std
 - possibility: learn σ_i for each feature independently

Similarity Graphs: Fully connected graphs

Edges connect everything.

- choose a “meaningful” similarity function s
- default choice:

$$s_{ij} = \exp \left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2} \right)$$

- why the exponential decay with the distance?
- σ controls the width of the neighborhoods
 - similar role as ε
 - a practical rule of thumb: 10% of the average empirical std
 - possibility: learn σ_i for each feature independently
- metric learning (a whole field of ML)

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold has its limits ($N \approx 10000$)*

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- *graph construction step can be a huge bottleneck*

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- *want to go higher? (we often have to)*

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing
 - ▶ **CoverTrees**

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing
 - ▶ **CoverTrees**
 - ▶ **Spectral sparsifiers**

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing
 - ▶ **CoverTrees**
 - ▶ **Spectral sparsifiers**
 - sometime we may not need the graph (just the final results)

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing
 - ▶ **CoverTrees**
 - ▶ **Spectral sparsifiers**
 - sometime we may not need the graph (just the final results)
 - yet another story: when we start with a large graph and want to make it sparse (later in the course)

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing
 - ▶ **CoverTrees**
 - ▶ **Spectral sparsifiers**
 - sometime we may not need the graph (just the final results)
 - yet another story: when we start with a large graph and want to make it sparse (later in the course)
- these rules have little theoretical underpinning

Similarity Graphs: Important considerations

- *calculate all s_{ij} and threshold* has its limits ($N \approx 10000$)
- graph construction step can be a huge bottleneck
- want to go higher? (we often have to)
 - down-sample
 - approximate NN
 - ▶ **LSH** - Locally Sensitive Hashing
 - ▶ **CoverTrees**
 - ▶ **Spectral sparsifiers**
 - sometime we may not need the graph (just the final results)
 - yet another story: when we start with a large graph and want to make it sparse (later in the course)
- these rules have little theoretical underpinning
- **similarity is very data-dependent**

Michal Valko

`michal.valko@inria.fr`

Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>